# UNIX® SYSTEM ADMINISTRATION HANDBOOK

EVI NEMETH
GARTH SNYDER · SCOTT SEEBASS

001 ServiceNow's Exhibit No. 1006

PRENTICE HALL SOFTWARE SERIES

# UNIX® System Administration Handbook

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**Prentice Hall**
Englewood Cliffs, New Jersey 07632

**Library of Congress Cataloging-in-Publication Data**

Nemeth, Evi. (date)
Unix system administration handbook / Evi Nemeth, Garth Snyder, Scott Seebass.
p. cm.
Includes index.
ISBN 0-13-933441-6
1. UNIX (Computer operating system) I. Snyder, Garth. II. Seebass, Scott. III. Title.
QA76.76.063N45 1989
005.4'3--dc20 89-32379
CIP

Editorial/production supervision: bookworks
Cartoons by: Tyler Stevens
Manufacturing buyer: Robert Anderson
UNIX® is a registered trademark of AT&T.
Prentice Hall Software Series, Brian W. Kernighan, *Advisor*

© 1989 by Prentice-Hall, Inc.
A Division of Simon & Schuster
Englewood Cliffs, New Jersey 07632

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College Technical and Reference Division
Englewood Cliffs, NJ 07632
(201) 592-2498

For information about our audio products, write us at:
Newbridge Book Clubs, 3000 Cindel Drive, Delran, NJ 08370

Printed in the United States of America
10 9 8 7 6 5

ISBN 0-13-933441-6

Prentice-Hall International (UK) Limited, London
Prentice-Hall of Australia Pty. Limited, Sydney
Prentice-Hall Canada Inc., Toronto
Prentice-Hall Hispanoamericana, S.A., Mexico
Prentice-Hall of India Private Limited, New Delhi
Prentice-Hall of Japan, Inc., Tokyo
Simon & Schuster Asia Pte. Ltd., Singapore
Editora Prentice-Hall do Brasil, Ltda., Rio de Janeiro

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# Preface

## Introduction

This book is about how to "do" UNIX system administration the right way. The techniques we will teach you are simple, efficient, and effective.

There are now over a million UNIX sites, and the pace at which new installations are added is steadily increasing. Computer vendors like UNIX because it is accessible and relatively easy to adapt to a new machine; the cost of porting UNIX is insignificant compared to the cost of writing a proprietary operating system. Computer buyers like UNIX because it's easy to find people who already know how to use it and because it allows them to create an expandable system without being tied to one particular vendor. Everybody wins.

UNIX is also a very elegant operating system. It is flexible, so that you never have the sense that you are fighting the system to force it into a mold it was never meant to fit. It is consistent, so that only one set of paradigms need be learned. It is powerful, so that the faculties of the system can be brought to bear against practically any problem.



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Unfortunately, UNIX is also cryptic, idiosyncratic, and unstandardized. Learning to use or administer UNIX requires a significant time investment. Once the initial period of frustration is over, however, you have a very powerful tool at your disposal.

Running a UNIX system is fun because you get to use the system for what it does best: writing, installing, and maintaining software. Since a high degree of automation can be grafted into UNIX administration, you will rarely find yourself repeating tiresome chores by hand. Just as in user-level UNIX, the standard paradigms, once learned, are clear and consistent.

**UNIX System Administration Handbook** is intended primarily for people with little or no background in system administration, but it is more than just a tutorial. Chapters are organized as much for reference as for straight-through reading. We hope that you'll be able to find what you need just by scanning the table of contents, but we've also taken pains to make the index accurate and thorough. We want this book to be indispensable.

We assume throughout that you are already familiar with the basics of UNIX and know how to use the shell and commands such as **mkdir**, **vi**, and **sh**. If you don't, you *must* supplement this book with some additional reading. Consult the UNIX bibliography at the end of this book for some specific recommendations.

## How this book differs from the UNIX manuals

Most UNIX systems come with enough documentation to kill a cow; you may wonder why a book about system administration is even necessary. The problem, as we see it, is that the standard documents describe various commands and subsystems *in isolation*, without providing a framework to tie them together. Manuals always tell you *how* to do things without ever telling you *why* you want to do them. And *why*, after all, is what system administration is all about.

UNIX machines used to be big and expensive, and the support personnel used to be specialists. Now you can buy a desktop UNIX system for $2000. The big-system paradigm has been shattered. We believe that as time goes on, UNIX administration will become less of an atomic specialty and more a component of many people's jobs.

Without guidance, it's easy for a new administrator to lose sight of ultimate goals in a sea of documentation or to inadvertently skip over one or more critical areas through failure to realize that they are important. That is where this book fits in. While we can't claim that *everything* you need to know is included, we can guarantee that after reading it you will know pretty much what UNIX administration involves and how to go about it.

## Hardware and software discussed

There are two main flavors of UNIX: AT&T's System V and the University of California at Berkeley's 4.*X* BSD (Berkeley Software Distribution). System V is oriented towards business, and so stability, simplicity, and maintainability are among its strengths. BSD embodies a number of advanced concepts not present in System V but is not as well supported. In past years BSD UNIX has maintained a technological lead over System V, but current trends in the System V world appear to be closing this gap.

Most of the computers at our site run some form of BSD, but some run System V variants and some run both at the same time. *The emphasis of this book is on BSD.* However, the two versions are often close enough so that by noting differences between the two we have produced a book that should be useful for either flavor.

Paradoxically, the differences among the various ports and versions of AT&T UNIX are sometimes much greater than the differences between BSD and "standard" System V. The information in this book regarding AT&T must therefore be taken with a grain of salt; when we say that something functions in a certain way under AT&T UNIX we mean that it is true for at least one major version and is usually true for several. Our closest reference machines are an AT&T 3B15 running System V.2.1.1 and an AT&T 3B2 running System V.3, but we've used a variety of other sources in trying to present a fair and balanced image of the AT&T world.

In this book, the letters "BSD" mean Berkeley UNIX 4.2, the version of BSD that most commercial ports are based on. We also cover Berkeley release 4.3, but everything specific to this revision is identified as such. Similarly, the letters "ATT" represent our homogenized view of AT&T UNIX.

Some machine-specific material has been introduced into this book in an effort to illustrate the variations among different flavors of UNIX. The four systems we've used as examples are:

- DEC Vaxen running pure BSD
- Sun Microsystems Sun3's running Sun UNIX
- Pyramid 90x's running dualPort OSx
- ATT 3B15 running System V.2

These are all nice systems; however, the fact that they appear in this book should not be construed as an endorsement. We've selected these four because they show some degree of variation from one another without being so widely separated that all comparisons are moot. Machine-specific examples are clearly marked.

Revisions 4.0 and higher of Sun's UNIX show significant deviations from BSD, especially with respect to the placement and format of system files. This book is still valid as a general reference for such systems; however, when we mention material specific to Suns in this book we are speaking mostly of revisions 3.2 and 3.4.

## Organization

It would be nice if operating systems were designed so that they could be explained in a linear way, without having to unravel a complex web of interrelations in order to create a coherent presentation. Alas, an operating system designed in this way would be easy to understand but probably not very useful. UNIX is especially horrible to explain because so little is built into the operating system itself; instead, much of the action shamelessly takes place in the context of user space—right in your living room, as it were.

Consequently, we've chosen an organization for this book that is chronological rather than conceptual. Rather than starting with the most basic facts and gradually building up to more complex abstractions, topics are introduced in the order that you would probably want to activate them in a new system. We like this organization because it tells you what you need to know *when you need to know it* and because we feel that it increases the value of each chapter as reference material. Except for two or three chapters that are fairly light, you will find the level of presentation fairly even.

The first six chapters of this book are presented in a sequence you should find useful if you are bringing up a system straight from the packing crate. In this section are contained the bare essentials, things you need to know just to get the system up and running. Chapters 7-14 introduce the various kinds of hardware that are typically used with UNIX and discuss how to install, configure, and operate this equipment. Chapters 15-19 discuss five important UNIX software subsystems: electronic mail, telephone communication, network news, backups, and accounting. The rest of the chapters represent a grab-bag of UNIX topics, some of which may not concern you.

## Attitude preadjustment

As a new UNIX system administrator, you are about to begin a long and complex project. As you work, keep in mind the five following rules. If you adhere to their spirit your success will be practically automatic.

### Be consistent

As long as you know exactly what you have done and how to undo it, you won't lose comprehension of the system you have created. The best way to maintain comprehension is to keep good documentation, but this is hard to do

consistently. The next best thing is to pick standard ways of performing system tasks, standard places to put various types of files, standard coding practices, etc., and stick to them.

### Learn more than you think you need to know

The more complete your comprehension of how your system works, the easier your life becomes. Time invested in learning about the system is paid back later many times over. In the long run, it will not save you time to learn the absolute minimum necessary for the completion of a particular job.

### Read the manuals

The manuals are the final authority on your system, and you should know their contents. This is not to say that you should read them from cover to cover — far from it! There is a sizable portion of the UNIX manuals that you may never have occasion to use. But when you learn a new command, you *should* skim its manual page. You will often find hidden features that are just right for your particular application.

### Do things right the first time

Don't patch something sloppily intending to come back later to fix it up. You won't. Make allowances for changes that might occur to your system; don't "hard-wire" system-dependent information into your programs and scripts. Make everything as bulletproof as possible. Assume Murphy's Law.

### Have fun

There is pleasure in doing things well, and this applies no less to computer administration than to any other activity. If you get frustrated or overstressed, work on something else for a while (if you can). The more good associations you develop with your administration duties, the better off you and your system will be.

## Send us bug reports

Please feel free to send suggestions, comments, and bug reports to us via electronic mail at sa-book@boulder.Colorado.EDU.

Good luck!

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